Section A Comprehensive Review - Mixed - Answer Key

1. $a_7 = 3$
2. $r = 2$; value = 720
3. $a_8 = 5$
4. $-2,\ 8,\ -32,\ 128,\ -512$
5. $a_5 = 64$
6. Yes; $r = 3$.
7. $r = 2$; next terms: $80,\ 160,\ 320$
8. $a_6 = 3072$
9. $r = 3$; next terms: $324,\ 972,\ 2916$
10. $r = \frac{1}{2}$; value = 150
11. $a_8 = 2187$
12. Yes; $r = -\frac{1}{2}$.
13. $r = \frac{1}{4}$; next terms: $4,\ 1,\ \frac{1}{4}$
14. $r = 2$; value = 960
15. $a_6 = 486$
16. $r = \frac{1}{2}$; value = 40
17. $r = \frac{1}{5}$; next terms: $1,\ \frac{1}{5},\ \frac{1}{25}$
18. No; the consecutive ratios are not constant.
19. $-5,\ -10,\ -20,\ -40,\ -80$
20. Yes; $r = 4$.
21. $a_7 = 5$
22. Yes; $r = -\frac{1}{2}$.
23. $a_6 = 4$
24. No; the consecutive ratios are not constant.